

Layered vs. Microservice Architecture — Case Study

Software Engineering (24PCA402) · Unit 08

Scenario

You are given a monolithic order-processing application (a single deployable with modules for catalog, orders, payments, and notifications, sharing one database).

Task 1 — Analyze the Monolith

Identify where modules are tightly coupled and where responsibilities overlap.

- Draw the current layered architecture (presentation / business / data)
- Mark points of high coupling and low cohesion

Task 2 — Propose a Microservice Split

Redesign the system as independently deployable services.

- Define service boundaries: Catalog, Orders, Payments, Notifications
- Specify the API contract between services
- Identify the new failure modes this introduces (network calls, data consistency)

Task 3 — Trade-off Analysis

Write a comparison covering deployability, scalability, operational complexity, and team ownership for both architectures.

Deliverable

Submit two architecture diagrams (before/after) and a one-page trade-off analysis.